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ORIGINAL ARTICLE



## Metabolic syndrome in menopause and associated factors: a meta-analysis

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### ABSTRACT

**Objective:** Metabolic syndrome (MetS) is a cluster of risk factors for cardiovascular disease and diabetes. Menopause is associated with an increased risk for MetS. The purpose of this meta-analysis is to better understand the relationship between MetS and menopause.

**Methods:** MEDLINE and EMBASE were searched for all the associated articles on (1) MetS components in postmenopausal women vs. premenopausal women, (2) comparison of MetS incidence between surgical menopause and natural menopause, (3) the effect of hormone therapy (HT) with 17 $\beta$ -estradiol (E2) compared to conjugated equine estrogen (CEE) on MetS components among postmenopausal women. A meta-analysis was applied by Review Manager 5.3 software.

**Results:** All comparable indicators were significantly unfavorably changed in postmenopausal women compared to premenopausal women except for high density lipoprotein cholesterol. Women who underwent surgical menopause suffered a 1.51-fold higher risk for MetS compared to those with natural menopause. HT with E2 provided more benefits for levels of triglyceride and diastolic blood, while CEE showed a better effect on both high and low density lipoprotein cholesterol levels.

**Conclusions:** Menopause nearly adversely affects all components of MetS, and surgical menopause may lead to a higher incidence of MetS compared to natural menopause. HT with various preparations may have different effects on MetS components. These results may clarify the management of menopause-related MetS in clinical practice.

### ARTICLE HISTORY

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### KEYWORDS

Menopause; metabolic syndrome; surgical menopause; hormone therapy

## Introduction

Menopause arises from exhaustion of the ovarian follicle pool<sup>1</sup>. It is accompanied by a significant decline in sex hormone levels, giving rise to distressing menopausal symptoms, including hot flushes, night sweats, mood disorders, sleep disturbances, urogenital disorders, bone loss and metabolic problems. Hormone therapy (HT) is the most effective treatment for women suffering from menopausal symptoms<sup>2</sup>.

Metabolic syndrome (MetS), previously known as insulin resistance syndrome or syndrome X, was first described by Reaven in 1988<sup>3</sup> and is a complex of interrelated risk factors for cardiovascular disease (CVD) and diabetes. These factors include insulin resistance, raised blood pressure, elevated triglyceride levels, low levels of high density lipoprotein (HDL) cholesterol, and obesity (particularly central adiposity)<sup>4</sup>. Data from the United States and Europe have reported the prevalence of MetS to be 20–25% of the adult population<sup>5,6</sup>. For postmenopausal women, the prevalence has been estimated to be significantly higher at 31–55%<sup>7–10</sup>. The type of menopause may impact on the risk. Some studies have found that surgical menopause results in a higher risk for MetS compared to natural menopause<sup>11–15</sup>. Any increase in MetS might seriously decrease quality of life amongst surgically postmenopausal women<sup>16</sup>.

As life expectancy increases, women will spend more than one-third of their lives postmenopausal. Good health in old age is critical for both individual women and the whole society. Studies have shown that estrogen deficiency is one of the key factors in the development of MetS in postmenopausal women<sup>17,18</sup>. HT use should therefore reduce the prevalence of postmenopausal MetS; however, the data on effects of HT remain somewhat controversial<sup>19–29</sup>.

We performed meta-analyses on components of MetS between pre- and postmenopause, MetS incidence between surgical and natural menopause, and the effect of HT on MetS components between natural hormones and non-body identical hormones.

## Methods

### Search strategy, inclusion and exclusion criteria

MEDLINE and EMBASE were searched up to July 31st, 2017 for all studies on:

- (1) MetS components in postmenopausal women vs. premenopausal women,
- (2) Comparison of MetS incidence between surgical menopause and natural menopause,

- (3) The effect of HT with 17 $\beta$ -estradiol (E2) compared to conjugated equine estrogen (CEE) on MetS components among postmenopausal women.

The references of all computer-identified publications were searched for additional studies, and the MEDLINE search for related articles was used to broaden the search. Review articles and references of other relevant studies identified were hand-searched to find additional eligible studies.

When comparing MetS components between postmenopausal and premenopausal women, keywords were combinations of 'menopause', 'premenopausal', 'postmenopausal', 'waist circumference', 'body mass index', 'triglycerides', 'high density lipoprotein cholesterol', 'low density lipoprotein cholesterol', 'fasting glucose', 'fasting insulin', 'blood pressure'.

Articles published in English were selected if they met all the following criteria:

- (1) Case-control or cohort studies, or randomized controlled trials;
- (2) Compared MetS components between post- and premenopausal women;
- (3) Data were extractable.

The exclusion criteria included participants who were known to be unhealthy or with confounding diseases, studies where data were duplicated in other publications or were not extractable.

When comparing MetS incidence between surgical menopause and natural menopause, keywords included 'surgical menopause', 'surgically menopausal', 'oophorectomy', 'natural menopause', 'metabolic syndrome', 'X syndrome'. Criteria for inclusion and exclusion were as stated above.

When examining the effect of HT with E2 versus CEE on MetS components among postmenopausal women, keywords included 'postmenopausal', '17 $\beta$ -estradiol', 'conjugated equine estrogen', 'waist circumference', 'body mass index', 'triglycerides', 'high density lipoprotein cholesterol', 'low density lipoprotein cholesterol', 'fasting glucose', 'fasting insulin', 'blood pressure'. Inclusion and exclusion criteria were as mentioned above.

### Data extraction

Data were extracted from eligible studies by two investigators, without blinding. For the first and third comparisons mentioned above, waist circumference, body mass index (BMI), triglycerides, HDL cholesterol, low density lipoprotein (LDL) cholesterol, fasting glucose, and fasting insulin were recorded by the investigators. When studying the MetS and surgical menopause, cases diagnosed with MetS in surgical and natural menopausal women were collected. All data were independently abstracted in duplicate by two researchers (D. P. and R. T.). If the investigators disagreed, a result was reached by discussion. For data not provided in table form or in the main text, the required information was obtained by contacting corresponding authors.

### Statistical analysis

Pooled analyses were estimated in the Review Manager 5.3 software (RevMan 5.3, The Nordic Cochrane Center, Rigshospitalet). Continuous variables were expressed as weighted mean difference (WMD) with 95% confidence intervals (CIs), while dichotomous data for each unit of analysis were expressed as an odds ratio (OR) with 95% CIs. The authors considered whether the clinical and methodological characteristics of the included studies were sufficiently similar for meta-analysis to provide a meaningful summary. The heterogeneity between studies was tested using the Q statistic. Statistical heterogeneity was assessed by the measure of  $I^2$ . An  $I^2$  measurement greater than 50% was taken to indicate a substantial heterogeneity<sup>30</sup>. If substantial heterogeneity was detected, a random effect model was used instead of a fixed effect model. A value of  $p < 0.05$  was considered statistically significant.

### Results

#### Meta-analysis of MetS components in postmenopausal vs. premenopausal women

A total of 706 articles were identified involving MetS components between pre- and postmenopausal women. Of these trials, 31 studies<sup>31–61</sup> met our inclusion criteria (Figure 1).

Pooled data on comparing components of MetS between premenopausal and postmenopausal women are shown in Table 1. Of the indicators feasible for meta-analysis, all results showed statistically significance ( $p < 0.01$ ) except for HDL cholesterol level ( $p = 0.72$ ).

Waist circumference and BMI were increased in postmenopausal women compared to premenopausal women (waist circumference: WMD 4.12, 95% CI 3.15–5.08; BMI: WMD 0.94, 95% CI 0.67–1.22).

Cardiovascular indicators including triglycerides (WMD 20.44, 95% CI 15.87–25.01), LDL cholesterol (WMD 20.77, 95% CI 14.61–26.94), systolic blood pressure (WMD 6.11, 95% CI 3.85–8.37) and diastolic blood pressure (WMD 3.54, 95% CI 2.14–4.93) were all adversely changed in postmenopausal women. Levels of fasting glucose (WMD 4.64, 95% CI 3.94–5.33) and fasting insulin (WMD 20.88, 95% CI 2.12–39.65) were both increased in the postmenopausal group (Figure 2).

#### Meta-analysis of MetS incidence in surgical menopause vs. natural menopause

A total of 79 articles comparing the prevalence of MetS between surgically and naturally menopausal women were identified. Only three of these<sup>14,15,62</sup> satisfied our criteria for inclusion (Figure 1). This left 428 surgically menopausal women and 1259 naturally menopausal women in the analysis. The incidence of MetS in surgically menopausal women was significantly higher than in naturally menopausal women (OR 1.51, 95% CI 1.21–1.90,  $p = 0.0003$ , Figure 3).

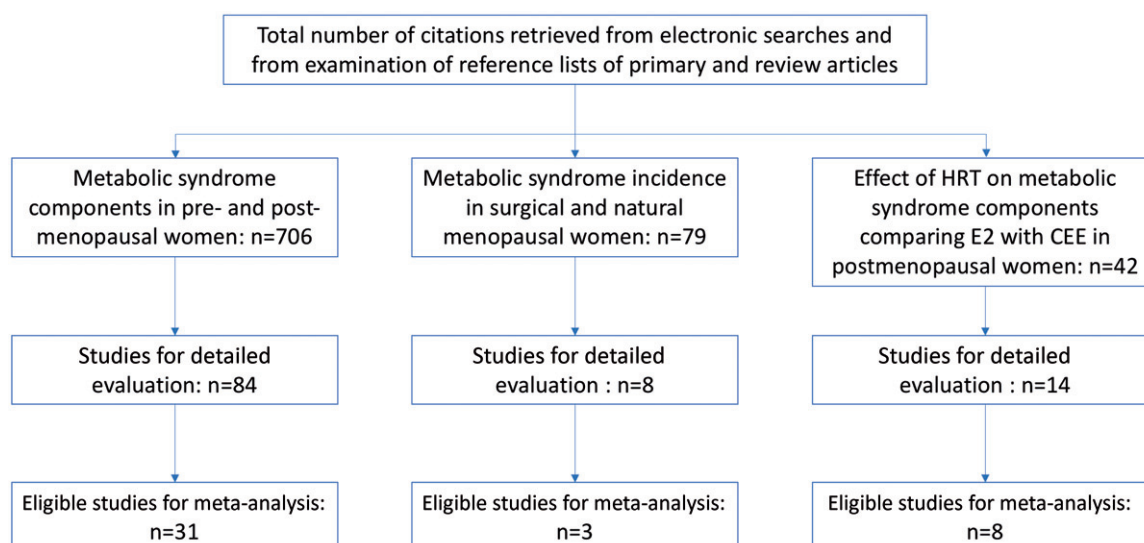


Figure 1. QUOROM statement flow diagram.

Table 1. Pooled results of metabolic syndrome components in postmenopausal vs. premenopausal women.

| Metabolic syndrome components        | Studies<br>n | Participants<br>n | Effect estimate<br>WMD (95% CI) | p-Value | Model  |
|--------------------------------------|--------------|-------------------|---------------------------------|---------|--------|
| Waist circumference (cm)             | 8            | 12277             | 4.12 (3.15–5.08)                | <0.0001 | Random |
| Body mass index (kg/m <sup>2</sup> ) | 21           | 14445             | 0.94 (0.67–1.22)                | <0.0001 | Random |
| Triglycerides (mg/dl)                | 19           | 9001              | 20.44 (15.87–25.01)             | <0.0001 | Random |
| HDL cholesterol (mg/dl)              | 19           | 13781             | 0.26 (–1.16–1.68)               | 0.72    | Random |
| LDL cholesterol (mg/dl)              | 17           | 13488             | 20.77 (14.61–26.94)             | <0.0001 | Random |
| Systolic blood pressure (mmHg)       | 12           | 1211              | 6.11 (3.85–8.37)                | <0.0001 | Random |
| Diastolic blood pressure (mmHg)      | 12           | 1211              | 3.54 (2.14–4.93)                | <0.0001 | Random |
| Glucose (mg/ml)                      | 15           | 12466             | 4.64 (3.94–5.33)                | <0.0001 | Random |
| Insulin (mmol/l)                     | 7            | 3023              | 20.88 (2.12–39.65)              | 0.03    | Random |

HDL, high density lipoprotein; LDL, low density lipoprotein; WMD, weighted mean difference; CI, confidence interval.

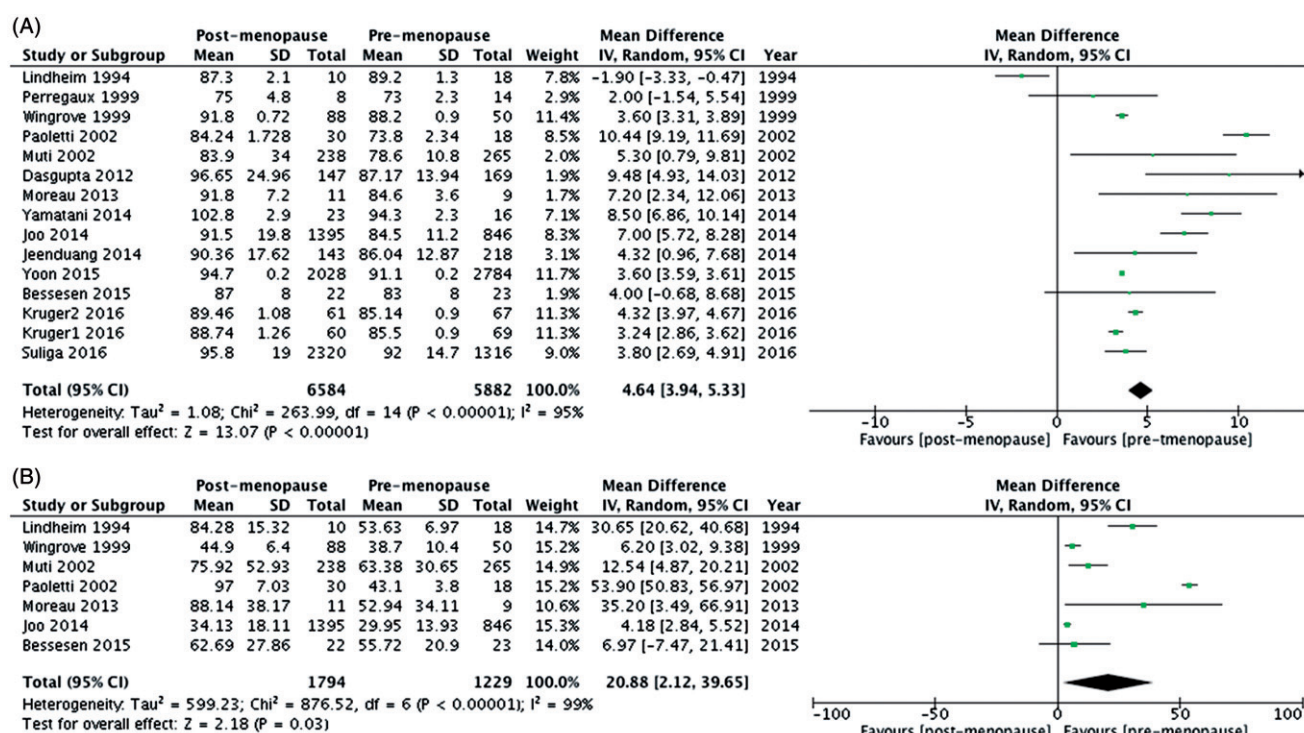


Figure 2. Meta-analysis of fasting glucose level (A) and fasting insulin level (B) in postmenopausal vs. premenopausal women.



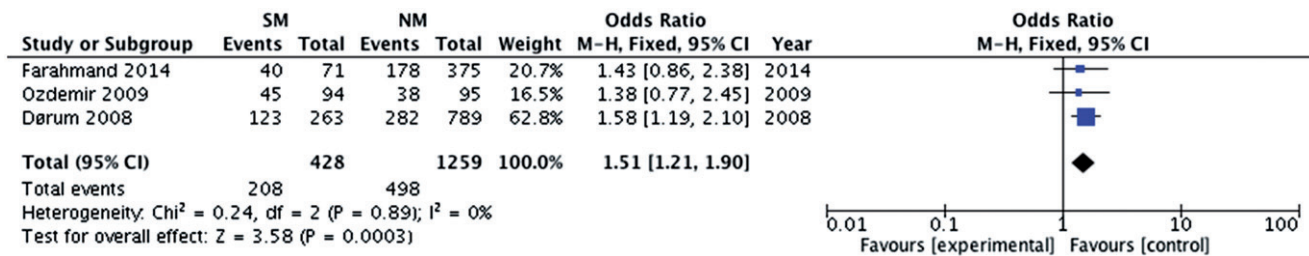


Figure 3. Meta-analysis of metabolic syndrome incidence between surgical menopause and natural menopause.

Table 2. Pooled results of hormone therapy with 17 $\beta$ -estradiol vs. conjugated equine estrogen on metabolic syndrome components before and after treatment in postmenopausal women.

| Metabolic syndrome components   | Studies<br>n | Participants<br>n | Effect estimate<br>WMD (95% CI) | p-Value | Model  |
|---------------------------------|--------------|-------------------|---------------------------------|---------|--------|
| <b>Baseline</b>                 |              |                   |                                 |         |        |
| Age (years)                     | 7            | 967               | 0.20 (−0.34, 0.74)              | 0.48    | Fixed  |
| Triglycerides (mg/dl)           | 7            | 811               | 2.12 (−4.65, 8.88)              | 0.54    | Fixed  |
| HDL cholesterol (mg/dl)         | 7            | 838               | −2.47 (−6.41, 1.47)             | 0.22    | Random |
| LDL cholesterol (mg/dl)         | 6            | 625               | 3.06 (−2.63, 8.75)              | 0.29    | Fixed  |
| Systolic blood pressure (mmHg)  | 4            | 169               | −2.32 (−6.30, 1.67)             | 0.25    | Fixed  |
| Diastolic blood pressure (mmHg) | 3            | 124               | −1.73 (−4.73, 1.27)             | 0.26    | Fixed  |
| <b>After treatment</b>          |              |                   |                                 |         |        |
| Triglycerides (mg/dl)           | 7            | 785               | −15.33 (−21.93, −8.73)          | <0.0001 | Fixed  |
| HDL cholesterol (mg/dl)         | 7            | 785               | −5.54 (−10.96, −0.13)           | 0.04    | Random |
| LDL cholesterol (mg/dl)         | 6            | 599               | 13.27 (8.04, 18.51)             | <0.0001 | Fixed  |
| Systolic blood pressure (mmHg)  | 4            | 161               | −3.19 (−7.22, 0.84)             | 0.12    | Fixed  |
| Diastolic blood pressure (mmHg) | 3            | 124               | −3.12 (−6.14, −0.10)            | 0.04    | Fixed  |

HDL, high density lipoprotein; LDL, low density lipoprotein; WMD, weighted mean difference; CI, confidence interval.

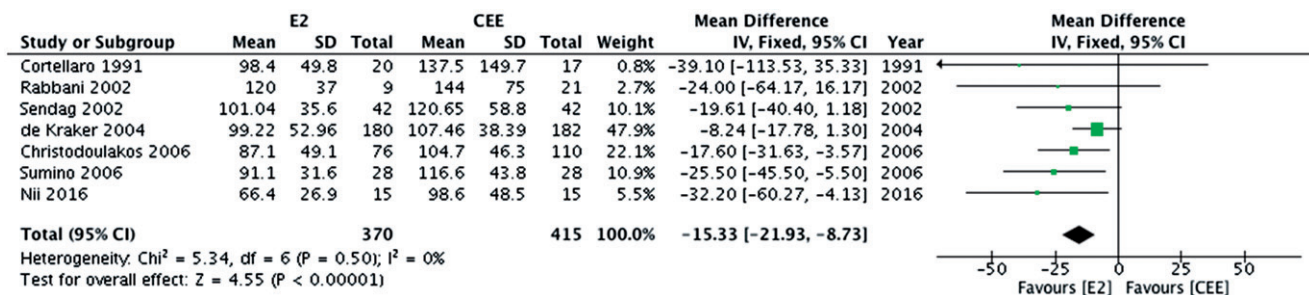


Figure 4. Meta-analysis of effect of hormone therapy with 17 $\beta$ -estradiol vs. conjugated equine estrogen on triglyceride levels in postmenopausal women.

### Meta-analysis of effect of HT with E2 vs. CEE on MetS components among postmenopausal women

A total of 42 trials were searched from electronic databases about the effect of HT with E2 and CEE on MetS components in postmenopausal women. Of these, only eight studies<sup>20–27</sup> met our inclusion criteria (Figure 1). Baseline data confirmed no difference between the two groups before treatment (Table 2). Participants received HT for at least 3 months. Compared to the CEE group, triglyceride levels (WMD 15.33, 95% CI 21.93 to −8.73, Figure 4) and diastolic blood pressure (WMD 3.12, 95% CI −6.14 to −0.10) levels were statistically lower in the E2 group. However, the level of HDL cholesterol (WMD 5.54, 95% CI −10.96 to −0.13) was higher in the CEE group, which also showed lower levels of LDL cholesterol (WMD 13.27, 95% CI 8.04–18.51). No difference was found in systolic blood pressure. Data on indicators for obesity and insulin resistance were not eligible for meta-analysis in this study.

### Discussion

#### MetS in premenopausal and postmenopausal women

MetS prevalence is raised after menopause. The prevalence of MetS is much lower among women than men before menopause<sup>63</sup>, but this situation is reversed after menopause<sup>64,65</sup>. Evidence shows that menopause is the independent risk factor for MetS regardless of age<sup>64,65</sup>, and it is believed that the decline in sex hormones, especially estrogen, is the likely etiology<sup>65,66</sup>. However, results from studies investigating MetS-associated indicators among women before and after menopause are not consistent. Our meta-analysis included 31 studies which analyzed detailed components of MetS in postmenopausal versus premenopausal women.

There have been several definitions of MetS since it was first defined by the World Health Organization (WHO) in 1998<sup>67</sup>. The definition proposed by the National Cholesterol

Education Program Adult Treatment Panel III<sup>68</sup> is commonly used in epidemiological studies. The definition proposed by the International Diabetes Federation (IDF) is mostly used in clinical practice<sup>69</sup>. This, in turn, has been updated in a Joint Interim Statement by the IDF Task Force on Epidemiology and Prevention in 2009<sup>4</sup>.

The criteria for women in the newest definition are as follows:

- (1) Elevated waist circumference (population- and country-specific definitions),
- (2) Elevated triglycerides ( $\geq 150$  mg/dl or on drug treatment),
- (3) Reduced HDL cholesterol level ( $< 50$  mg/dl or on drug treatment),
- (4) Elevated blood pressure (systolic blood pressure  $\geq 130$  mmHg and/or diastolic blood pressure  $\geq 85$  mmHg or antihypertensive drug treatment),
- (5) Elevated fasting glucose ( $\geq 100$  mg/dl) or on drug treatment.

It is generally agreed that abdominal obesity should not be a prerequisite for diagnosis and consequently it has been agreed that the presence of any three of five risk factors constitutes a diagnosis of MetS. We have sought to analyze all the parameters mentioned above in this meta-analysis.

Abdominal obesity is regarded as the critical factor of MetS<sup>70</sup>. Obesity occurs nearly three to five times more often in postmenopausal women than in premenopausal women<sup>71,72</sup>. Consistent with more obesity, BMI was  $0.94 \text{ kg/m}^2$  and waist circumference  $4.12 \text{ cm}$ , respectively, higher in postmenopausal women compared to premenopausal women.

Overweight or obese women had a greater risk of MetS compared to those with a normal weight<sup>73</sup>. The mechanisms for the increase in body fat, particularly mean central obesity, during menopause are still not fully elucidated. The most obvious reason is the decline in estrogen level, which is responsible for the accumulation of fat in the subcutaneous tissue, especially in the gluteal and femoral regions<sup>73</sup>.

Lipid metabolism dysfunction is another important component in the development of MetS and is also a major cause of CVD<sup>74</sup>. The risk of CVD is significantly increased after menopause regardless of age<sup>75</sup>.

In this meta-analysis, we found that levels of triglycerides and LDL cholesterol rose in postmenopausal women. Triglycerides were increased by a mean  $20.44 \text{ mg/dl}$  and LDL cholesterol by a mean  $20.77 \text{ mg/dl}$ . No statistical difference was detected in HDL cholesterol level ( $p = 0.72$ ). Some studies have reported that menopause is associated with an increase in total cholesterol, LDL cholesterol, triglycerides, and decrease in HDL cholesterol<sup>39,53,75</sup>; others have shown no effects on these indicators<sup>32,43,52,54</sup>, whilst two reported an increase in levels of HDL cholesterol<sup>56,57</sup>. The conclusion of this meta-analysis is that menopause is associated with dyslipidemia, especially regarding triglycerides and LDL cholesterol.

Together with the aging process, hypertension is also a major risk factor for cardiovascular morbidity and mortality in

postmenopausal women. The prevalence of hypertension in women older than 65 years is estimated to be about 60%<sup>76</sup>. This meta-analysis found a rise in both systolic and diastolic blood pressure ( $6.11 \text{ mmHg}$  for systolic blood pressure and  $3.54 \text{ mmHg}$  for diastolic blood pressure) in postmenopausal women, consistent with the results from Staessen and colleagues<sup>77</sup>. Postmenopausal women also show higher carotid-femoral pulse wave velocity, even after adjustment for age and BMI, which may explain the greater rise in systolic blood pressure after menopause<sup>78</sup>. A sudden decrease in estrogen level is another possible causative factor for the adverse changes in blood pressure<sup>78</sup>.

Insulin resistance is a critical component of MetS. We found that fasting glucose and insulin levels were increased dramatically in postmenopausal women ( $4.64 \text{ mg/dl}$  and  $20.88 \text{ mmol/l}$  higher compared to premenopausal women, respectively). All but two of the studies examined<sup>33,48</sup> found increased fasting glucose and insulin among postmenopausal women. Previous studies have shown that the risk for diabetes is five times higher in individuals with MetS<sup>79,80</sup>. Moreover, MetS with insulin resistance, as opposed to MetS without insulin resistance, has an additive effect of up to six- to seven-fold increased risk of type 2 diabetes<sup>81</sup>.

In this meta-analysis, our findings suggested that almost all the MetS-associated components (BMI, waist circumference, LDL cholesterol, triglycerides, systolic blood pressure, diastolic blood pressure, fasting glucose and insulin) except for HDL cholesterol were unfavorably changed after menopause. These variations may directly cause metabolic dysfunction and a higher risk for diabetes and CVD. Two studies explored the timing of metabolic change in relation to the menopause<sup>82,83</sup> and both found that the higher prevalence of MetS in postmenopausal women was more likely triggered by changes during the menopausal transition than by postmenopausal status.

### *Surgical menopause has a higher risk for MetS*

The age at natural menopause occurs most commonly around age 50 years<sup>11</sup>, although slightly earlier in Chinese women<sup>84</sup>. Surgically menopausal women usually experience the menopause at an earlier age and suffer from a greater prevalence and severity of menopausal symptoms. Pooled results from this meta-analysis showed that individuals with surgical menopause had a 1.51-fold raised risk for MetS compared to naturally menopausal women. The opportunistic removal of ovaries at the time of hysterectomy for benign conditions is also increasing the incidence of surgical menopause, especially amongst those women approaching menopausal age<sup>85–88</sup>. Surgically induced menopause results in a sudden decrease of sex hormones and this is considered the sole cause of the greater prevalence and severity of menopausal symptoms and related disease. Nevertheless, uncertainty remains regarding the true effects of surgical versus natural menopause<sup>89–92</sup> and more studies, especially prospective studies, are needed to help us understand the influence of surgical menopause on metabolism.

### Effect of HT with E2 vs. CEE on MetS components

There is no doubt that HT is the most effective treatment for menopause-related symptoms and disease. HT has experienced several controversies in recent years<sup>93</sup>. However, newly updated guidelines on HT use have clearly shown that HT, when prescribed to healthy postmenopausal women within 10 years of their last menstrual period, is a safe and effective option. More evidence-based information is also available regarding appropriate indications, doses and duration of therapy, thus maximizing benefits and minimizing risk<sup>2</sup>. The role of HT in the management of postmenopausal MetS has been debated for more than 10 years<sup>94,95</sup>. An earlier meta-analysis in 2006<sup>96</sup> found that HT reduces abdominal obesity, insulin resistance, new-onset diabetes, lipids and blood pressure in women without diabetes and reduced insulin resistance and fasting glucose in women with diabetes.

Newer studies have focused on safety of hormone preparations<sup>28,29</sup>. Published data suggested a safer profile for HT containing 17 $\beta$ -estradiol and natural progesterone compared with HT containing CEE and synthetic progestins<sup>97</sup>. 17 $\beta$ -Estradiol is the human-derived natural estrogen, while CEE is a natural formulation of an extract prepared from pregnant mares' urine. Both are commonly prescribed in clinical practice. In this meta-analysis, we found that, amongst women statistically similar at baseline, the use of HT for at least 3 months resulted in a significantly greater reduction of triglyceride levels and diastolic blood pressure in those women treated with E2 compared to those treated with CEE. However, there was a favorable change in HDL cholesterol and LDL cholesterol concentration with CEE treatment compared to E2. It is important to note that these results may have been influenced by different formulations of progestins used in these studies.

### Limitations

There are several limitations to our meta-analysis. The first part of the meta-analysis made comparisons between MetS components in pre- and postmenopausal women. However, age ranges were not totally consistent, which may have affected our results.

When examining the effects of surgical menopause on MetS, only three studies satisfied our criteria and no prospective studies have been published, thus limiting the strength of our findings.

When comparing the effects of E2 and CEE on the indicators of MetS, we were unable to correct for the various progestogens used. As synthetic progestins compared to natural progesterone or no progestogen at all are known to affect parameters of cardiovascular health, we cannot be sure that the results that we report are due entirely to different estrogenic effects or to other factors.

### Conclusions

Menopause is an important risk factor for metabolic syndrome. Most of the individual components of metabolic syndrome are unfavorably modified after menopause. Surgical

menopause with a sudden drop of ovarian function at an earlier age may further increase the risk for metabolic syndrome.

Hormone therapy appears to reduce the risk of both the components of metabolic syndrome and overall metabolic syndrome. The major benefit appears to reside with estrogen. Whether there is a difference in the overall effects of estradiol compared to conjugated estrogens remains unclear. Recent published data suggest that the beneficial effects of estrogens on metabolic syndrome are, to some extent, attenuated by synthetic progestins and less so by natural progesterone. Further high-quality clinical and basic researches are necessary to illustrate the exact mechanism and establish more helpful management for menopause-related metabolic syndrome.

**Conflict of interest** All authors report no conflict of interest. The authors alone are responsible for the content and writing of this paper.

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### References

1. Shifren JL, Gass ML. NAMS Recommendations for Clinical Care of Midlife Women Working Group. The North American Menopause Society recommendations for clinical care of midlife women. *Menopause* 2014;21:1038–62
2. Baber RJ, Panay N, Fenton A. IMS Writing Group. 2016 IMS Recommendations on women's midlife health and menopause hormone therapy. *Climacteric* 2016;19:109–50
3. Reaven GM. Banting lecture 1988. Role of insulin resistance in human disease. *Diabetes* 1988;37:1595–607
4. Alberti KG, Eckel RH, Grundy SM, et al. Harmonizing the metabolic syndrome: a joint interim statement of the International Diabetes Federation Task Force on Epidemiology and Prevention; National Heart, Lung, and Blood Institute; American Heart Association; World Heart Federation; International Atherosclerosis Society; and International Association for the Study of Obesity. *Circulation* 2009;120:1640–5
5. Ford ES, Giles WH, Dietz WH. Prevalence of the metabolic syndrome among US adults: findings from the third National Health and Nutrition Examination Survey. *JAMA* 2002;287:356–9
6. Beltrán-Sánchez H, Harhay MO, Harhay MM, et al. Prevalence and trends of metabolic syndrome in the adult U.S. population, 1999–2010. *J Am Coll Cardiol* 2013;62:697–703
7. Stefanska A, Bergmann K, Sypniewska G. Metabolic syndrome and menopause: pathophysiology, clinical and diagnostic significance. *Adv Clin Chem* 2015;72:1–75
8. Park YW, Zhu S, Palaniappan L, et al. The metabolic syndrome: prevalence and associated risk factor findings in the US population from the Third National Health and Nutrition Examination Survey, 1988–1994. *Arch Intern Med* 2003;163:427–36
9. Pucci G, Alcidì R, Tap L, et al. Sex- and gender-related prevalence, cardiovascular risk and therapeutic approach in metabolic syndrome: a review of the literature. *Pharmacol Res* 2017;120:34–42
10. Shah D. The annual conference of the British menopause society: June 2010. *J Midlife Health* 2010;1:48–50
11. Farahmand M, Ramezani Tehrani F, Bahri Khomami M, et al. Surgical menopause versus natural menopause and cardio-metabolic disturbances: a 12-year population-based cohort study. *J Endocrinol Invest* 2015;38:761–7
12. Michelsen TM, Pripp AH, Tonstad S, et al. Metabolic syndrome after risk-reducing salpingo-oophorectomy in women at high risk



- for hereditary breast ovarian cancer: a controlled observational study. *Eur J Cancer* 2009;45:82–9
13. Kilic S, Yilmaz N, Erdogan G, et al. Effect of non-oral estrogen on risk markers for metabolic syndrome in early surgically menopausal women. *Climacteric* 2010;13:55–62
  14. Farahmand M, Ramezani Tehrani F, Simbar M, Mehrabi Y, Khalili D, Azizi F. Does metabolic syndrome or its components differ in naturally and surgically menopausal women? *Climacteric* 2014;17:348–55
  15. Ozdemir S, Celik C, Gorkemli H, et al. Compared effects of surgical and natural menopause on climacteric symptoms, osteoporosis, and metabolic syndrome. *Int J Gynaecol Obstet* 2009;106:57–61
  16. Kahn R, Buse J, Ferrannini E, et al. The metabolic syndrome: time for a critical appraisal. Joint statement from the American Diabetes Association and the European Association for the Study of Diabetes. *Diabetologia* 2005;48:1684–99
  17. Coyoy A, Guerra-Araiza C, Camacho-Arroyo I. Metabolism regulation by estrogens and their receptors in the central nervous system before and after menopause. *Horm Metab Res* 2016;48:489–96
  18. Morselli E, Santos RS, Criollo A, et al. The effects of oestrogens and their receptors on cardiometabolic health. *Nat Rev Endocrinol* 2017;13:352–64
  19. Kim C, Halter JB. Endogenous sex hormones, metabolic syndrome, and diabetes in men and women. *Curr Cardiol Rep* 2014;16:467
  20. Nii S, Shinohara K, Matsushita H, et al. Hepatic effects of estrogen on plasma distribution of small dense low-density lipoprotein and free radical production in postmenopausal women. *J Atheroscler Thromb* 2016;23:810–18
  21. Christodoulakos GE, Lambrinoudaki IV, Economou EV, et al. Differential effect of hormone therapy and tibolone on lipids, lipoproteins, and the atherogenic index of plasma. *J Cardiovasc Pharmacol* 2006;47:542–8
  22. Cortellaro M, Nencioni T, Boschetti C, et al. Cyclic hormonal replacement therapy after the menopause: transdermal versus oral treatment. *Eur J Clin Pharmacol* 1991;41:555–9
  23. de Kraker AT, Kenemans P, Smolders RG, et al. The effects of 17 beta-oestradiol plus dydrogesterone compared with conjugated equine oestrogens plus medroxyprogesterone acetate on lipids, apolipoproteins and lipoprotein(a). *Maturitas* 2004;49:253–63
  24. Ichikawa J, Sumino H, Ichikawa S, et al. Different effects of transdermal and oral hormone replacement therapy on the renin-angiotensin system, plasma bradykinin level, and blood pressure of normotensive postmenopausal women. *Am J Hypertens* 2006;19:744–9
  25. Rabbani LE, Seminario NA, Sciacca RR, et al. Oral conjugated equine estrogen increases plasma von Willebrand factor in postmenopausal women. *J Am Coll Cardiol* 2002;40:1991–9
  26. Sendag F, Karadas N, Ozsener S, et al. Effects of sequential combined transdermal and oral hormone replacement therapies on serum lipid and lipoproteins in postmenopausal women. *Arch Gynecol Obstet* 2002;266:38–43
  27. Sumino H, Ichikawa S, Kasama S, et al. Different effects of oral conjugated estrogen and transdermal estradiol on arterial stiffness and vascular inflammatory markers in postmenopausal women. *Atherosclerosis* 2006;189:436–42
  28. Farr JN, Khosla S, Miyabara Y, et al. Effects of estrogen with micronized progesterone on cortical and trabecular bone mass and microstructure in recently postmenopausal women. *J Clin Endocrinol Metab* 2013;98:E249–57
  29. Wroolie TE, Kenna HA, Williams KE, et al. Differences in verbal memory performance in postmenopausal women receiving hormone therapy: 17 $\beta$ -estradiol versus conjugated equine estrogens. *Am J Geriatr Psychiatry* 2011;19:792–802
  30. The Cochrane Collaboration. Cochrane Handbook for Systematic reviews of interventions, Version 5.1.0. Available from: <http://handbook-5-1.cochrane.org/> [last accessed 2011]
  31. Bednarek-Tupikowska G, Filus A, Kuliczowska-Plaksej J, et al. Serum leptin concentrations in pre- and postmenopausal women on sex hormone therapy. *Gynecol Endocrinol* 2006;22:207–12
  32. Behall KM, Scholfield DJ, Hallfrisch J. Whole-grain diets reduce blood pressure in mildly hypercholesterolemic men and women. *J Am Diet Assoc* 2006;106:1445–9
  33. Bessesen DH, Cox-York KA, Hernandez TL, et al. Postprandial triglycerides and adipose tissue storage of dietary fatty acids: impact of menopause and estradiol. *Obesity (Silver Spring)* 2015;23:145–53
  34. Blumenthal JA, Matthews K, Fredrikson M, et al. Effects of exercise training on cardiovascular function and plasma lipid, lipoprotein, and apolipoprotein concentrations in premenopausal and postmenopausal women. *Arterioscler Thromb* 1991;11:912–17
  35. Campbell MJ, Woodside JV, Honour JW, et al. Effect of red clover-derived isoflavone supplementation on insulin-like growth factor, lipid and antioxidant status in healthy female volunteers: a pilot study. *Eur J Clin Nutr* 2004;58:173–9
  36. Dasgupta S, Salman M, Lokesh S, et al. Menopause versus aging: the predictor of obesity and metabolic aberrations among menopausal women of Karnataka, South India. *J Midlife Health* 2012;3:24–30
  37. Farag NH, Whitsett TL, McKey BS, et al. Caffeine and blood pressure response: sex, age, and hormonal status. *J Womens Health (Larchmt)* 2010;19:1171–6
  38. Grandjean PW, Crouse SF, O'Brien BC, et al. The effects of menopausal status and exercise training on serum lipids and the activities of intravascular enzymes related to lipid transport. *Metabolism* 1998;47:377–83
  39. Joo JK, Hong GP, Han SE, et al. The association between serum uric acid level and incidence of metabolic syndrome according to menopausal status in Korean women. *J Menopausal Med* 2014;20:126–32
  40. Kruger MC, Chan YM, Kuhn-Sherlock B, et al. Differential effects of calcium- and vitamin D-fortified milk with FOS-inulin compared to regular milk, on bone biomarkers in Chinese pre- and postmenopausal women. *Eur J Nutr* 2016;55:1911–21
  41. Lindheim SR, Legro RS, Bernstein L, et al. Behavioral stress responses in premenopausal and postmenopausal women and the effects of estrogen. *Am J Obstet Gynecol* 1992;167:1831–6
  42. Lindheim SR, Buchanan TA, Duffy DM, et al. Comparison of estimates of insulin sensitivity in pre- and postmenopausal women using the insulin tolerance test and the frequently sampled intravenous glucose tolerance test. *J Soc Gynecol Investig* 1994;1:150–4
  43. Manabe E, Aoyagi K, Tachibana H, et al. Relationship of intra-abdominal adiposity and peripheral fat distribution to lipid metabolism in an island population in western Japan: gender differences and effect of menopause. *Tohoku J Exp Med* 1999;188:189–202
  44. Menzel J, Di Giuseppe R, Biemann R, et al. Association between omentin-1, adiponectin and bone health under consideration of osteoprotegerin as possible mediator. *J Endocrinol Invest* 2016;39:1347–55
  45. Moreau KL, Deane KD, Meditz AL, et al. Tumor necrosis factor- $\alpha$  inhibition improves endothelial function and decreases arterial stiffness in estrogen-deficient postmenopausal women. *Atherosclerosis* 2013;230:390–6
  46. Muti P, Quattrin T, Grant BJ, et al. Fasting glucose is a risk factor for breast cancer: a prospective study. *Cancer Epidemiol Biomarkers Prev* 2002;11:1361–8
  47. Paoletti AM, Pilloni M, Orrù M, et al. Efficacy and safety of oral and transdermal hormonal replacement treatment containing levonorgestrel. *Maturitas* 2002;42:137–47
  48. Perregaux D, Chaudhuri A, Mohanty P, et al. Effect of gender differences and estrogen replacement therapy on vascular reactivity. *Metabolism* 1999;48:227–32
  49. Prelevic GM, Beljic T. The effect of oestrogen and progestogen replacement therapy on systolic flow velocity in healthy postmenopausal women. *Maturitas* 1994;20:37–44
  50. Sanada M, Nakagawa H, Kodama I, et al. The effect of hormone replacement therapy on metabolism of lipoprotein remnants in postmenopausal women. *Maturitas* 2000;34:75–82
  51. Schaefer EJ, Lamon-Fava S, Cohn SD, et al. Effects of age, gender, and menopausal status on plasma low density lipoprotein



- cholesterol and apolipoprotein B levels in the Framingham Offspring Study. *J Lipid Res* 1994;35:779–92
52. Stoney CM, West SG, Hughes JW, *et al.* Acute psychological stress reduces plasma triglyceride clearance. *Psychophysiology* 2002;39: 80–5
  53. Suliga E, Koziół D, Cieśla E, *et al.* Factors associated with adiposity, lipid profile disorders and the metabolic syndrome occurrence in premenopausal and postmenopausal women. *PLoS One* 2016;11:e0154511
  54. Ushiroyama T, Nosaka S, Ueki M. Short-term effects of low-dose atorvastatin on inflammatory status and lipid profiles in perimenopausal hypercholesterolemic, hypertriglyceridemic women. *Int J Cardiol* 2006;113:66–75
  55. van Beek AP, de Ruijter-Heijstek FC, Erkelens DW, *et al.* Menopause is associated with reduced protection from postprandial lipemia. *Arterioscler Thromb Vasc Biol* 1999;19:2737–41
  56. Vega-López S, Vidal-Quintanar RL, Fernandez ML. Sex and hormonal status influence plasma lipid responses to psyllium. *Am J Clin Nutr* 2001;74:435–41
  57. Wingrove CS, Garr E, Pickar JH, *et al.* Effects of equine oestrogens on markers of vasoactive function in human coronary artery endothelial cells. *Mol Cell Endocrinol* 1999;150:33–7
  58. Yamatani H, Takahashi K, Yoshida T, *et al.* Differences in the fatty acid metabolism of visceral adipose tissue in postmenopausal women. *Menopause* 2014;21:170–6
  59. Yoon BI, Han KD, Lee KW, *et al.* Insulin resistance is associated with prevalence of physician-diagnosed urinary incontinence in postmenopausal non-diabetic adult women: data from the Fourth Korea National Health and Nutrition Examination Survey. *PLoS One* 2015;10:e0141720
  60. Zhang C, Zhuang Y, Qiang H, *et al.* Relationship between endogenous estrogen concentrations and serum cholesteryl ester transfer protein concentrations in Chinese women. *Clin Chim Acta* 2001;314:77–83
  61. Jeenduang N, Trongsakul R, Inhomgsa P, *et al.* The prevalence of metabolic syndrome in premenopausal and postmenopausal women in Southern Thailand. *Gynecol Endocrinol* 2014;30:573–6
  62. Dørum A, Tonstad S, Liavaag AH, *et al.* Bilateral oophorectomy before 50 years of age is significantly associated with the metabolic syndrome and Framingham risk score: a controlled, population-based study (HUNT-2). *Gynecol Oncol* 2008;109:377–83
  63. Zuo H, Shi Z, Hu X, *et al.* Prevalence of metabolic syndrome and factors associated with its components in Chinese adults. *Metabolism* 2009;58:1102–8
  64. Ben Ali S, Belfki-Benali H, Aounallah-Skhiri H, *et al.* Menopause and metabolic syndrome in tunisian women. *Biomed Res Int* 2014;2014: 457131
  65. Eshtiaghi R, Esteghamati A, Nakhjavani M. Menopause is an independent predictor of metabolic syndrome in Iranian women. *Maturitas* 2010;65:262–6
  66. Sharma S, Aggarwal N, Joshi B, *et al.* Prevalence of metabolic syndrome in pre- and post-menopausal women: a prospective study from apex institute of North India. *J Midlife Health* 2016;7:169–74
  67. Alberti KG, Zimmet PZ. Definition, diagnosis and classification of diabetes mellitus and its complications. 1. Diagnosis and classification of diabetes mellitus. Provisional report of a WHO consultation. *Diabet Med* 1998;15:539–53
  68. Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults. Executive Summary of The Third Report of The National Cholesterol Education Program (NCEP) expert panel on detection, evaluation, and treatment of high blood cholesterol in adults (Adult Treatment Panel III). *JAMA* 2001;285:2486–97
  69. Alberti KG, Zimmet P, Shaw J. IDF Epidemiology Task Force Consensus Group. The metabolic syndrome—a new worldwide definition. *Lancet* 2005;366:1059–62
  70. Rusinek H, Convit A. Obesity: cerebral damage in obesity-associated metabolic syndrome. *Nat Rev Endocrinol* 2014;10:642–4
  71. Donato GB, Fuchs SC, Oppermann K, *et al.* Association between menopause status and central adiposity measured at different cutoffs of waist circumference and waist-to-hip ratio. *Menopause* 2006;13:280–5
  72. Kwaśniewska M, Pikala M, Kaczmarczyk-Chałas K, *et al.* Smoking status, the menopausal transition, and metabolic syndrome in women. *Menopause* 2012;19:194–200
  73. St-Onge MP, Janssen I, Heymsfield SB. Metabolic syndrome in normal-weight Americans: new definition of the metabolically obese, normal-weight individual. *Diabetes Care* 2004;27:2222–8
  74. McQueen MJ, Hawken S, Wang X, *et al.* Lipids, lipoproteins, and apolipoproteins as risk markers of myocardial infarction in 52 countries (the INTERHEART study): a case-control study. *Lancet* 2008;372:224–33
  75. Stokes J, 3rd, Kannel WB, Wolf PA, *et al.* The relative importance of selected risk factors for various manifestations of cardiovascular disease among men and women from 35 to 64 years old: 30 years of follow-up in the Framingham Study. *Circulation* 1987;75:V65–73
  76. Taddei S. Blood pressure through aging and menopause. *Climacteric* 2009;12(Suppl 1):36–40
  77. Staessen J, Bulpitt CJ, Fagard R, *et al.* The influence of menopause on blood pressure. *J Hum Hypertens* 1989;3:427–33
  78. Coylewright M, Reckelhoff JF, Ouyang P. Menopause and hypertension: an age-old debate. *Hypertension* 2008;51:952–9
  79. Grundy SM. Metabolic syndrome: connecting and reconciling cardiovascular and diabetes worlds. *J Am Coll Cardiol* 2006;47: 1093–100
  80. Ford ES, Li C, Sattar N. Metabolic syndrome and incident diabetes: current state of the evidence. *Diabetes Care* 2008;31:1898–904
  81. Meigs JB, Rutter MK, Sullivan LM, *et al.* Impact of insulin resistance on risk of type 2 diabetes and cardiovascular disease in people with metabolic syndrome. *Diabetes Care* 2007;30:1219–25
  82. Gurka MJ, Vishnu A, Santen RJ, *et al.* Progression of metabolic syndrome severity during the menopausal transition. *J Am Heart Assoc* 2016;5:e003609
  83. Zivkovic TB, Vuksanovic M, Jelic MA, *et al.* Obesity and metabolic syndrome during the menopause transition in Serbian women. *Climacteric* 2011;14:643–8
  84. Li L, Wu J, Pu D, *et al.* Factors associated with the age of natural menopause and menopausal symptoms in Chinese women. *Maturitas* 2012;73:354–60
  85. Hendrix SL. Bilateral oophorectomy and premature menopause. *Am J Med* 2005;118(Suppl 12B):131–5
  86. Shoupe D, Parker WH, Broder MS, *et al.* Elective oophorectomy for benign gynecological disorders. *Menopause* 2007;14:580–5
  87. Laughlin GA, Barrett-Connor E, Kritz-Silverstein D, *et al.* Hysterectomy, oophorectomy, and endogenous sex hormone levels in older women: the Rancho Bernardo Study. *J Clin Endocrinol Metab* 2000;85:645–51
  88. Howard BV, Kuller L, Langer R, *et al.* Risk of cardiovascular disease by hysterectomy status, with and without oophorectomy: the Women's Health Initiative Observational Study. *Circulation* 2005;111:1462–70
  89. Lobo RA. Surgical menopause and cardiovascular risks. *Menopause* 2007;14:562–6
  90. Parker WH, Broder MS, Liu Z, *et al.* Ovarian conservation at the time of hysterectomy for benign disease. *Clin Obstet Gynecol* 2007;50:354–61
  91. Bonithon-Kopp C, Scarabin PY, Darne B, *et al.* Menopause-related changes in lipoproteins and some other cardiovascular risk factors. *Int J Epidemiol* 1990;19:42–8
  92. Tuna V, Alkış I, Safiye AS, Imamoglu N, Bayram N, Ali Ismet T. Variations in blood lipid profile, thrombotic system, arterial elasticity and psychosexual parameters in the cases of surgical and natural menopause. *Aust NZ J Obstet Gynaecol* 2010;50:194–9
  93. Boardman HM, Hartley L, Eisinga A, *et al.* Hormone therapy for preventing cardiovascular disease in post-menopausal women. *Cochrane Database Syst Rev* 2015;19:CD002229

94. Saltevo J, Puolakka J, Ylikorkala O. Plasma endothelin in postmenopausal women with type 2 diabetes mellitus and metabolic syndrome: a comparison of oral combined and transdermal oestrogen-only replacement therapy. *Diabetes Obes Metab* 2000;2:293–8
95. Hsia J, Bittner V, Tripputi M, et al. Metabolic syndrome and coronary angiographic disease progression: the Women's Angiographic Vitamin & Estrogen trial. *Am Heart J* 2003;146:439–45
96. Salpeter SR, Walsh JM, Ormiston TM, et al. Meta-analysis: effect of hormone-replacement therapy on components of the metabolic syndrome in postmenopausal women. *Diabetes Obes Metab* 2006;8:538–54
97. Mirkin S, Amadio JM, Bernick BA, et al. 17 $\beta$ -Estradiol and natural progesterone for menopausal hormone therapy: REPLENISH phase 3 study design of a combination capsule and evidence review. *Maturitas* 2015;81:28–35